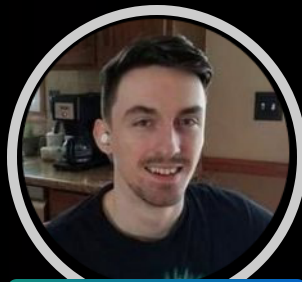


JOSH BEDO

FULL-STACK ENGINEER



Made with HTML/CSS

ERIE PA, US

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Native English, US citizen

github.com/joshbedo

LANGUAGES

 JavaScript	18 yrs
 TypeScript	5 yrs
 HTML / CSS	18 yrs
 Go	3 yrs
 Ruby	7 yrs
 PHP	5 yrs
 Python	3 yrs
 Rust	1 yr

 = I've been using it since it came out

FRONTEND

 React	
 SolidJS	2 yrs
 Next.js	
 Tailwind	
 Styled components	
 Redux	
 React Query	
 TRPC	
 Webpack	
 Babel	

BACKEND

 Node.js	
 Ruby on Rails	7 yrs
 PHP	5 yrs
 Python	3 yrs
 Elixir	< 1 yr
 Step Functions	2 yrs
 RabbitMQ	5 yrs
 Monolithic	5 yrs
 Microservices	5 yrs
 REST APIs	15 yrs

DEV OPS

 Git	10 yrs	 App Sync	
 Docker	5 yrs	 AWS CDK	5 yrs
 ECS / EKS	~ 6 yrs	 Cloudformation	5 yrs
 EC2 / Fargate		 Cognito	2 yrs
 DynamoDB		 Lambda	
 API Gateway		 New Relic	5 yrs

Hey there 🙋 I'm Josh, I've been designing and building software since I was 15. At 18, I moved to New York City and spent the next 10 years working across a mix of startups and enterprise companies. Along the way, I contributed to fast-moving teams at companies like Namely (Series D, acquired), and Reserve (backed by Google Ventures and Uber, later acquired by Resy / American Express).

BEST AT

- Pixel Perfect Design
- Crafting User Experiences
- Problem Solving
- Listening
- Simplifying and communicating complex ideas

HOBBIES

- Drawing and Painting
- Playing Guitar & Drums
- Cooking
- Golf
- Going to concerts
- Playing Magic the Gathering or board game nights with friends

INTERESTED IN

- Designing scalable platforms that blend performance, usability, and maintainability
- Experimenting with and learning new technologies
- Building cool stuff with cool people
- Using data-driven decisions to build products users want and enjoy using
- Using my art & design background to craft thoughtful and beautiful experiences
- Building tools that improve developer speed, clarity, and happiness

EXPERIENCE

Portland, OR (Remote) — 06/2025 – 12/2025

Knapsack Senior Software Engineer

PROBLEM: SEARCH PLATFORM RE-ARCHITECTURE

Legacy search was expensive, slow, and returned noisy / irrelevant results, making it difficult for users to find components and tokens across large design systems.

SOLUTION

- Migrated search to full-text indexing with redesigned data models and serialization.
- Implemented Redis caching and scheduled index-sync cron jobs to keep results consistent across environments.
- Rebuilt frontend + backend search pipelines, improving relevance scoring and reducing over-fetching.

IMPACT

- Reduced irrelevant search results by ~60%
- Cut search-related infra cost by ~35%
- Improved median search response time by ~2x

PROBLEM: AI PLATFORM / MCP SERVER ARCHITECTURE

Existing APIs were not designed to support LLM-driven generation of UI from design system tokens or AI-enriched component metadata.

SOLUTION

- Helped design and implemented an MCP server that ingests customer tokens + components to generate production-ready React pages.
- Built a pipeline that extracts structured metadata from React component libraries, enriches it via AI/ML, and serves it at scale.
- Extended ingestion workflows and redesigned token normalization to support AI pipelines across microservices.

IMPACT

- Enabled launch of AI-powered page generation + metadata enrichment features.
- Cut time from concept to live web page by ~70%
- Created scalable foundation for future AI features across the platform.

PROBLEM: FIGMA → TOKEN → CODE PIPELINE

Tokens extracted from Figma were inconsistently serialized, leading to incomplete or inaccurate LLM-generated code.

SOLUTION

- Rebuilt Figma token extraction APIs with improved normalization for colors, typography, spacing, other tokens and components.
- Fixed serialization/deserialization edge cases across services to ensure token fidelity in downstream prompts.

IMPACT

- Improved generated React code accuracy and reduced token mismatch bugs by ~50%

PROBLEM: PLATFORM MODERNIZATION (NEXT.JS 13 → 16)

Dashboard performance degraded due to heavy CSR and outdated build tooling.

SOLUTION

- Led migration of the dashboard and supporting services to Next.js 16.
- Shifted critical paths from CSR → SSR and improved bundling strategy.

IMPACT

- Reduced TTI (Time to Interactive) by ~40%
- Improved build times by ~30% across environments

PROBLEM: CUSTOMIZABLE NAVIGATION BUILDER

Customers couldn't flexibly organize design system navigation across side-nav, top-nav, and secondary menus.

SOLUTION

- Re-architected navigation into a drag-and-drop tree system using dnd-kit.
- Replaced deprecated drag-and-drop dependencies that were blocking the Next.js 16 upgrade.
- Built a customizable nav editor enabling per-customer hierarchy configuration.

IMPACT

- Unblocked enterprise clients managing 1k+ components per system.
- Reduced customer support requests related to navigation by ~45%

TypeScript

Next.js

Node.js

Go

Docker

GraphQL

Cypress

Playwright

Meilisearch

Datadog

Redis

Postgres

MCP

Figma API

RTK Query

AWS

Mountain View, California - 03/2024 - 05/2025

Udacity
Software Engineer

- Developed reusable and accessible React components using Chakra UI, with full RTL localization support, reducing UI-related bugs and boosting developer efficiency across teams.
- Integrated A/B and multivariate testing using Optimizely and Segment, enabling rapid experimentation and data-driven product decisions.
- Contributed to the development of scalable classroom features used by individual learners, enterprise clients, and government partners.
- Led development of AI-powered classroom features including video transcript syncing, integrated coding workspaces, and intelligent help desk support.
- Built an AI chatbot for the classroom that answered course-specific questions, quoted lesson content, and collected structured feedback from learners.
- Developed integrations with cloud-based learning environments (AWS, Azure, GCP, VMWare, OpenAI), enabling hands-on labs and auto-provisioned resources for student courses.
- Collaborated on the implementation of an AI-based skill tracking system that guided students in setting goals and identifying relevant courses to achieve them.
- Engineered drag-and-drop quiz components and other performance-testing features within a shared design system, increasing component reuse and consistency.
- Collaborated across teams in a large-scale microservice ecosystem with over 100 micro-frontends and microservices, using technologies like React, Node.js, Golang, Python, Ruby on Rails, PHP, and Bun.

TypeScriptNext.jsReact QueryRTK QueryGraphQLStorybookPlaywrightCypressGolang

Ruby on RailsPHPOpenAIAnthropicSegment

Pittsburgh, PA - 01/2020 - Present

Apptivity
Software Consultant

- Led development of Appraisal.com, an automotive valuation platform built with Laravel, enabling users to assess real-time car, real estate, gold, and other asset values based on live market data.
- Developed a Home Value Forecaster that visualized historical and projected real estate trends, including income per household, population density by ZIP code, average property values, and year-over-year pricing changes.
- Built a B2B product research app in PHP for Shopify merchants, allowing users to analyze market trends, identify product opportunities, and connect with verified vendors for fulfillment.
- Contributed to a PHP-based social network for artists, designed to help musicians discover collaborators, share projects, and connect within a creative community.
- Built a custom Learning Management System (LMS) using PHP and WordPress to support virtual classrooms, enabling teachers and students to manage coursework, assignments, and communication.
- Contracted as a full-time engineer for EcoATM/Gazelle, Post.News, Sony, and Universal Studios

TypeScriptReactReact NativeNode.jsAWSGraphQLPostgresRedisPHPWordpress

ShopifyDocker

EXPERIENCE

New York, NY - 10/2022 - 05/2023

Post.news

Frontend Developer

- Established a comprehensive component design system utilizing Storybook. This system significantly improved our ability to reuse components across various features, streamlining development, and enhancing consistency across the app.
- Collaborated closely with the engineering and design team to help build out new features and migrate existing Next.js/XState features over to SolidJS. Their Next.js app was having really bad performance issues with XState and the architecture wasn't scaling so we switched to SolidJS.
- Collaborated with the engineering and product team to setup an Analytics event tracking system that we reused throughout our monorepo across different apps to track various user metrics across Amplitude and Google Tag Manager/Analytics

TypeScript

Next.js

XState

SolidJS

Vite

Storybook

TurboRepo

Cognito

Fargate

Datadog

San Diego, CA - 07/2021 - 10/2022

EcoATM / Gazelle

Senior Full-Stack Engineer

- Designed and implemented scalable APIs using GraphQL/AppSync and multiple in-house and third-party APIs using AWS Lambda, Step Functions and other AWS technologies that serve millions of IMEI device check requests daily and support device trade-ins.
- Developed systems to help maintain and enhance the device trade-in system and CMS used for editing products and page content using AWS Lambda, Step Functions, React, Ruby on Rails, Cognito, Event Bridge, AWS Fargate, Postgres, New Relic, Webpack, RSpec, CircleCI, Shopify Liquid, Contentful
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TypeScript

React

Rails

AppSync

Lambda

Step Functions

Event Bridge

Cognito

New York, NY - 10/2018 - 04/2019

Ralph Interactive

Full-Stack Engineer

- Crafted micro experience applications for a diverse client base in the entertainment industry including Netflix, Nickelodeon, Comedy Central, USA Network, Sony, and others utilizing a wide-range of technologies.
- Developed a custom experience for the Bandersnatch episode of the Netflix series Black Mirror
- Developed a character generator for the Netflix series Super Drags where you can create your own drag queen. We used PHP and Vue.js for this project.
- Created a Chrome Extension for the Netflix series Queer Eye
- Developed a PHP website for the USA Network series Real Country that featured interactive elements like an emoji poll and emojis for video timelines
- Developed a Facebook Messenger bot using PHP and the Facebook API for the Netflix series Miracle Workers. The messenger bot walks visitors through a job application experience applying to be a miracle worker and work with god.
- Developed a marketing page for Sony's artist revenue tracking platform.

React

Preact

Rails

PHP

Vue.js

Three.js

SASS

Anime.js

Chrome Extensions

EXPERIENCE

New York, NY - 12/2015 - 01/2017

LiveWith

Senior Full-Stack Engineer / Co-founder

- Designed and developed a mobile app using React Native to help people moving find roommates and apartments that matched their combined preferences.
- Developed a weighted preference matching feature using Elasticsearch to do a full-text search and find roommates or apartments that fit your criteria.
- Designed and developed a tinder-like UI for matching with potential roommates and apartments.
- Developed nightly cron job to pull in apartment listings from our third parties to keep our listings fresh and up to date.
- Developed marketing landing pages and integrated analytics and pixels for tracking marketing efforts this helped us gain our first 100 users.

React Native

Node.js

Elasticsearch

AWS Lambda

Firebase

Docker

Microservices

New York, NY - 02/2016 - 09/2016

The Daily Beast

Senior Full-Stack Engineer

- During the 2016 election our traffic spiked to 120k requests per second causing our site to go down. I helped implement an article caching solution using Redis and Nginx to cache articles and serve them to users.
- Collaborated with the product design team and developed a React UI component library containing the various components we reused through the application.
- Developed a CMS system for writers to draft and publish content. Before they were sharing documents so this was a huge upgrade and made collaboration easier.
- Collaborated with the analytics team to integrate Optimizely and other tracking for various PPC third parties like Outbrain, Taboola, and others.
- Developed an infinite scrolling solution for news articles that increased pages visited, time spent on pages, and increased PPC ad revenue.

React

Jest

Node.js

Redis

Nginx

Docker

AWS ECS

AWS DynamoDB

New York, NY - 08/2015 - 01/2016

Reserve

Senior Core Backend Engineer

- Developed microservice APIs with Node.js/Sails.js to support locking in restaurant reservations across web and mobile using Redis and Websockets.
- After Resy was acquired helped develop restaurant table management software so restaurants can map out their floor layout for different times of the day.
- Helped with engineering efforts for migrating the Ruby on Rails table management software over to more event-driven Node.JS APIs since most of our features needed to work in real-time through websockets. ex: booking a reservation
- Developed the payments service or refunds, partial refunds, etc. using third-party service Braintree and Node.js

React

Node.js

Sails.js

Microservices

Rails

PHP

Redis

Postgres

RabbitMQ

EXPERIENCE

New York, NY - 01/2015 - 08/2015

Namely

Senior Full-Stack Engineer

- Developed features for an enterprise HR company supporting onboarding, payroll, employee performance measurement, and social feed using Ruby on Rails, CoffeeScript, Backbone, React, and Postgres.
- Collaborated with the product design team to develop a React UI styleguide with unit and E2E tests.
- Developed a React upgrade tool using Python to upgrade from React 0.10.x to 0.12.x. This swapped out deprecated methods like `.createClass` with new ones and helped us upgrade our codebase.
- As an HR company we had an onboarding process for companies once they joined the platform. I developed features for exporting, importing, and organizing other company documents using S3/Cloudfront.
- Added unit/integration testing for React during Backbone.js migration using mocha, chai, sinon, and jsdom.

React

Backbone.js

Rails

PHP

Postgres

Python

Go

Cloudfront

Mocha

Chai

Sinon

New York, NY - 01/2015 - 08/2015

PINCHme

Full-Stack Engineer

- Developed features for a high traffic eCommerce site that gre to 1m users the first year I worked there.
- Collaborated with the product team and redesigned the flow for surveys/reviews which increased the member engagement by 20%
- Developed the rewards page displaying user points and achievements using Ruby on Rails, Backbone, and Marionette
- Developed a gamification system to replace our third-party Gigya. This system rewarded users for completing various actions like watching a video, leaving a review, completing a survey, etc.
- Improved our build system by shifting over to CommonJS modules instead of storing everything globally on `window.app.models`, `window.app.controllers`, etc.
- Load tested and optimized queries with n+1 lookups using tools like JMeter, Fiddler, and New Relic
- Developed and maintained unit and integration tests using Mocha, Chai, Sinon, RSpec, Cucumber, and Selenium

Backbone

Marionette

Rails

Selenium

Cucumber

RSpec

Mocha

JMeter

New Relic

Jersey City, NJ - 08/2012 - 02/2013

JPMorgan Chase

Software Engineer

- Developed enterprise features for JPMorgan websites using vanilla JavaScript, jQuery, and Java
- Introduced newer CSS3 features with fallback support for IE7 and IE8. Shared findings with the team about performance trade-offs for supporting older browser with polyfills.
- Developed JavaScript methods to handle accessibility features on screen readers.
- Developed new auth pages for chase.com so we could reuse elements across JPMorgan websites.
- Developed various page load performance optimizations to improve page load times.

Vanilla JavaScript

jQuery

Java

Tomcat

* Shortened because I was told my resume was too long 😊